

SMART GRID STABILIZATION & PREDICTION SYSTEM

Table - 30

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Problem Statement

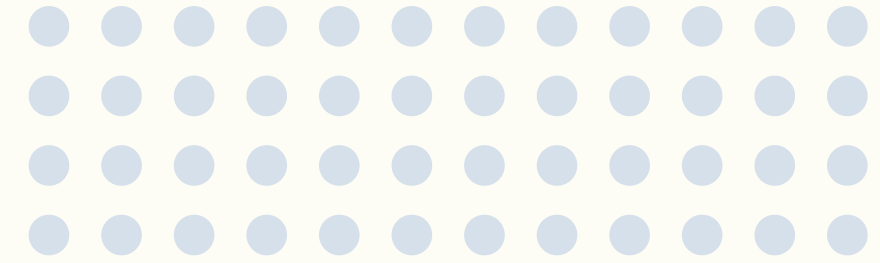
With the increasing adoption of electric vehicles (EVs), power grids are experiencing significant stress due to unplanned and simultaneous charging. This leads to:

- Grid overloads and blackouts during peak hours
- Long wait times at overcrowded charging stations
- Higher electricity costs due to inefficient charging schedules

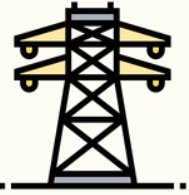


There is a need for an intelligent system that can predict, optimize, and balance EV charging demand to ensure grid stability, reduce wait times, and improve energy efficiency.

Proposed Solution



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GRID STABILITY PREDICTION

Uses AI to forecast grid load and prevent overloads.



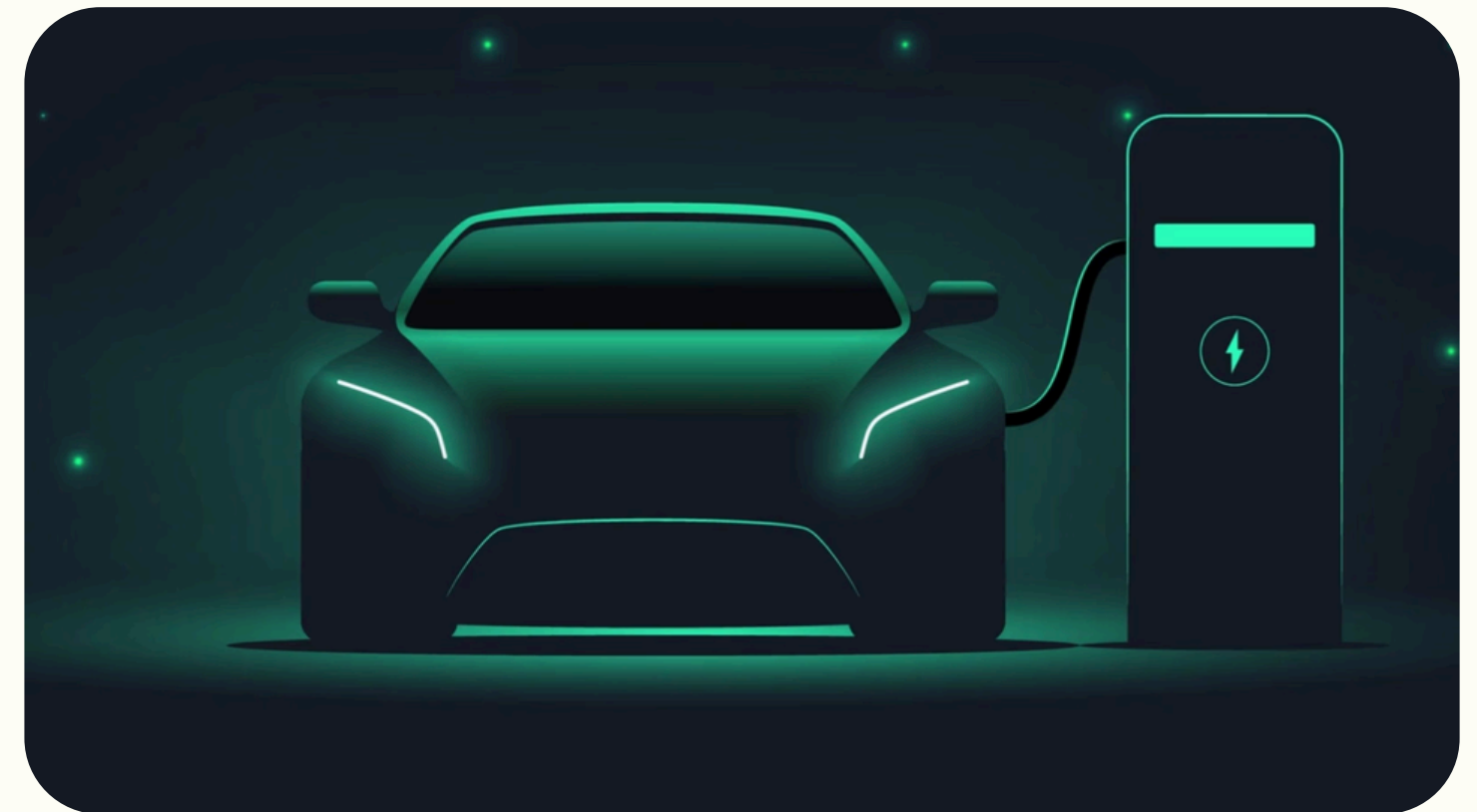
PEAK HOURS PREDICTION

Predicts peak hours, and helps charging stations to recharge during non peak hours

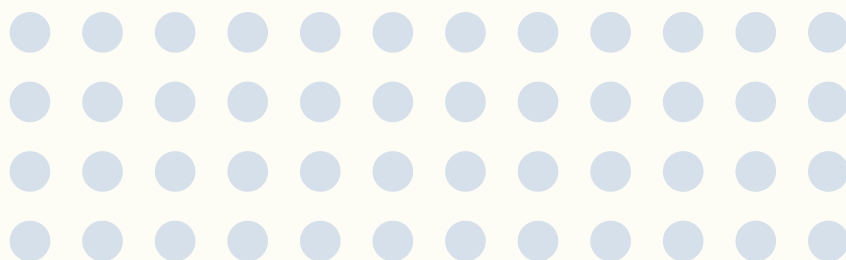


USER RECOMMENDATIONS

Provides real-time suggestions for charging locations and times via an app or dashboard.

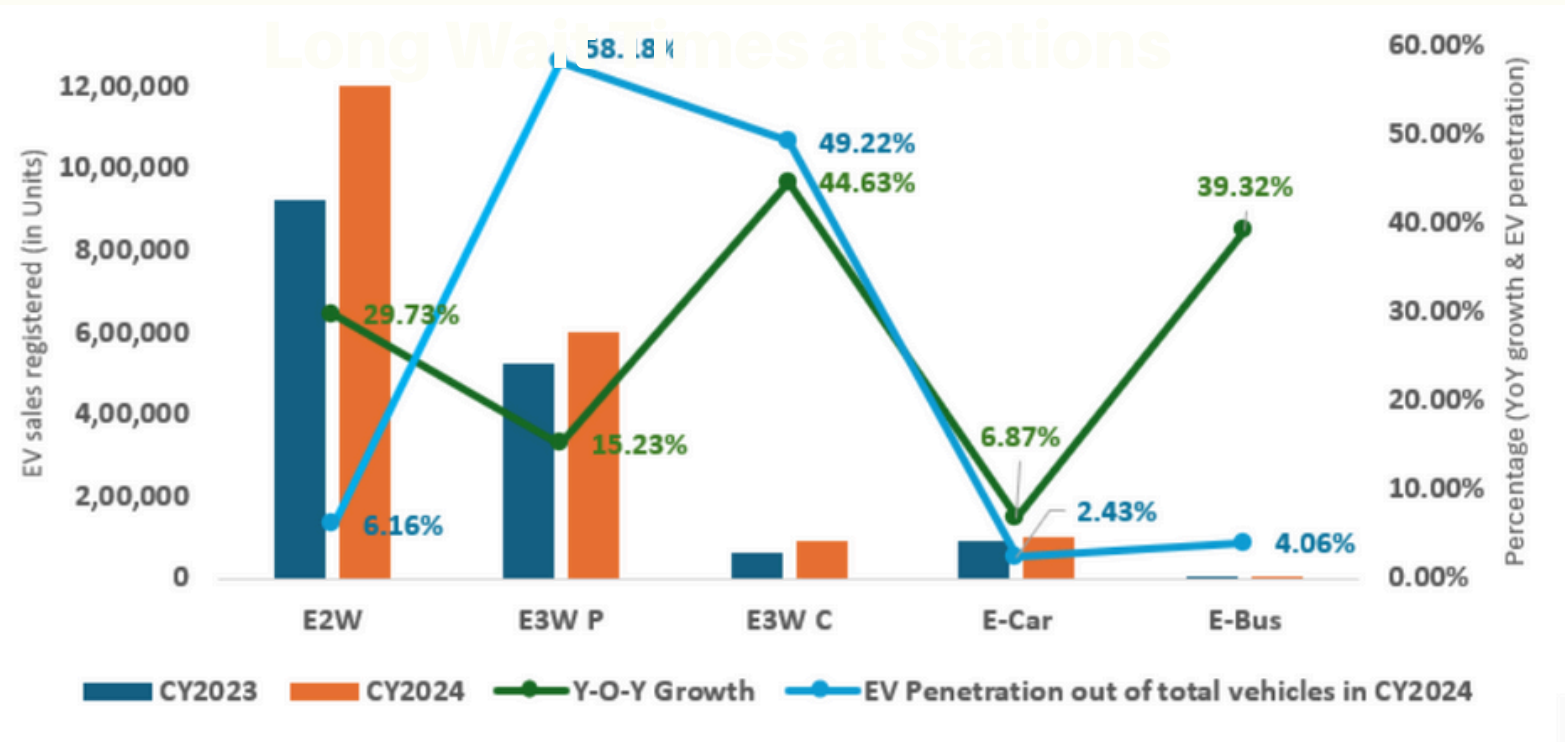


Market Research



Increased EV sales - EV BOOM

There will be a surge in EV in upcoming years.



Over half of all new cars sold in the U.S. by 2030 are expected to be electric vehicles



A Major Strain on Power grid

Domestic electricity demand in 2022 is expected to increase up to 18% by 2030 and 38% by 2035

That's a big change over the roughly 5% increase we saw in the past decade.



Software Stack

Frontend

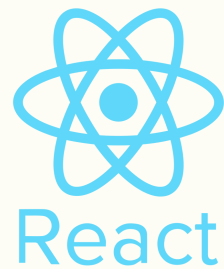


Chart.js

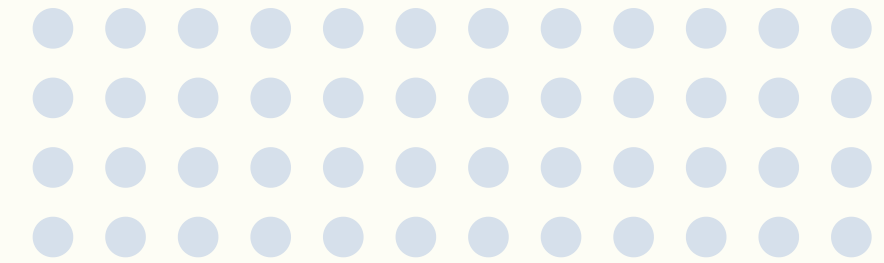
Responsive UI: Ensures seamless experience across devices.

Backend



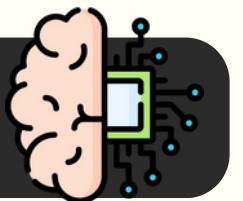
Flask

RESTful APIs: Facilitates seamless data exchange between components



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Machine Learning



PyTorch

Analysis: Develop AI models for charging predictions and grid optimization.

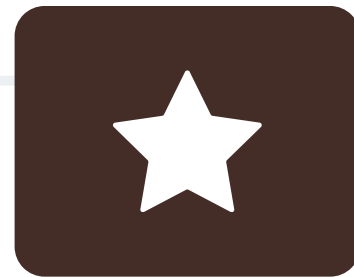
RELEVANCE TO EVS & AUTOMOBILES

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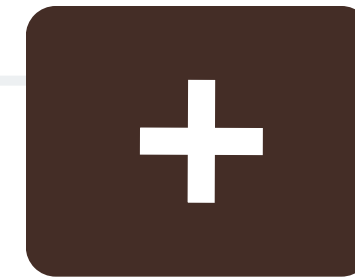
FOR EV OWNERS

Saves time, reduces charging cost



FOR POWER COMPANIES

Prevents grid failures, balances demand



FOR CHARGING OPERATORS

Increases station utilization & revenue



FOR GOVERNMENTS

Promotes clean energy usage,
Promotes V2G by incentivization.



Existing Solutions

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Time-of-Use Pricing (Dynamic Charging)

Ford's Dynamic Charging is supported by Octopus Energy in the U.K., allowing EVs to charge during optimal times based on renewable energy availability and cost



SECO's Load Management

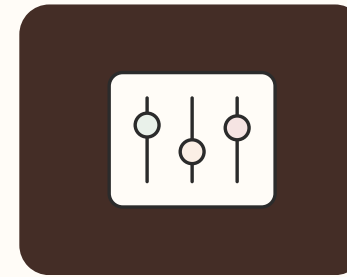
SECO's Edge AI charging station management software employs predictive AI technology to detect maintenance needs, manage charging station operations, and provide real-time service status to customers¹



Real Time Grid Management

EVs can be incentivized to charge or discharge based on grid conditions, helping stabilize the energy supply.

Expected Impacts



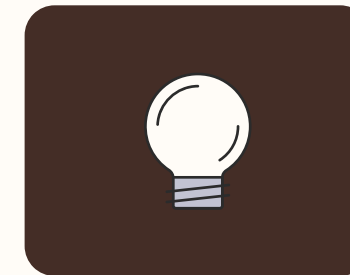
Reduced Blackouts

Reduces blackouts by preventing grid overload



Decreased Wait times

Decreases wait times at charging stations



Electricity Cost

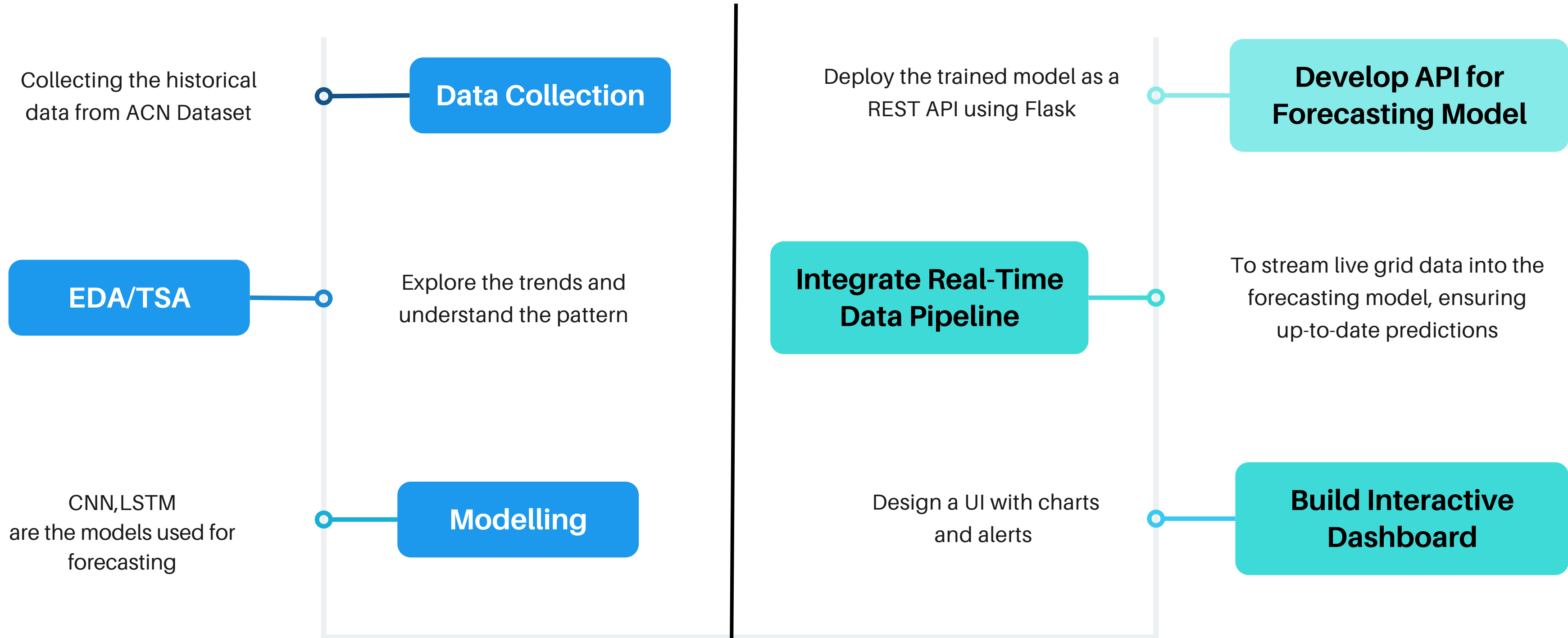
Saves 20-30% electricity costs for EV users by incentivization and

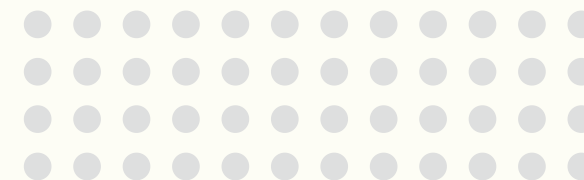
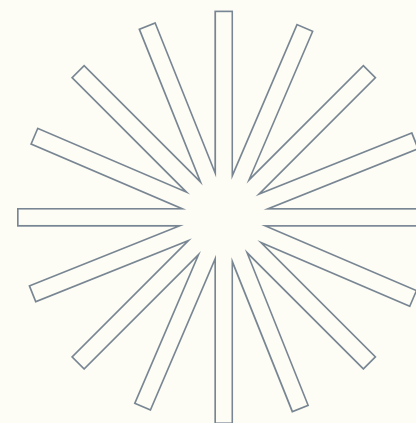


Overall Integrity

Improves overall EV infrastructure planning

ACTION PLAN





THANK YOU!

